Geochemical Evolution of Neoproterozoic Volcanic Series, South China

Table 4. Major (wt%) and trace element (ppm) analytical dataset of volcanic rocks.

Sample	Unit	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	Rb	Sr	Y	Zr	Nb	Ba
VOL-01	Lower	65.38	0.37	14.62	3.29	1.45	2.18	3.86	4.27	98.0	278.0	18.9	169.0	12.7	662
VOL-02	Lower	65.56	0.39	14.74	3.38	1.50	2.26	3.92	4.34	101.0	276.0	19.3	173.0	13.0	674
VOL-03	Lower	65.74	0.41	14.86	3.47	1.55	2.34	3.98	4.41	104.0	274.0	19.7	177.0	13.3	686
VOL-04	Lower	65.92	0.43	14.98	3.56	1.60	2.42	4.04	4.48	107.0	272.0	20.1	181.0	13.6	698
VOL-05	Lower	66.10	0.45	15.10	3.65	1.65	2.50	4.10	4.55	110.0	270.0	20.5	185.0	13.9	710
VOL-06	Lower	66.28	0.47	15.22	3.74	1.70	2.58	4.16	4.62	113.0	268.0	20.9	189.0	14.2	722
VOL-07	Lower	66.46	0.49	15.34	3.83	1.75	2.66	4.22	4.69	116.0	266.0	21.3	193.0	14.5	734
VOL-08	Lower	66.64	0.51	15.46	3.92	1.80	2.74	4.28	4.76	119.0	264.0	21.7	197.0	14.8	746
VOL-09	Lower	66.82	0.53	15.58	4.01	1.85	2.82	4.34	4.83	122.0	262.0	22.1	201.0	15.1	758
VOL-10	Lower	67.00	0.55	15.70	4.10	1.90	2.90	4.40	4.90	125.0	260.0	22.5	205.0	15.4	770
VOL-11	Lower	67.18	0.57	15.82	4.19	1.95	2.98	4.46	4.97	128.0	258.0	22.9	209.0	15.7	782
VOL-12	Lower	67.36	0.59	15.94	4.28	2.00	3.06	4.52	5.04	131.0	256.0	23.3	213.0	16.0	794
VOL-13	Lower	67.54	0.61	16.06	4.37	2.05	3.14	4.58	5.11	134.0	254.0	23.7	217.0	16.3	806
VOL-14	Lower	67.72	0.63	16.18	4.46	2.10	3.22	4.64	5.18	137.0	252.0	24.1	221.0	16.6	818
VOL-15	Lower	67.90	0.65	16.30	4.55	2.15	3.30	4.70	5.25	140.0	250.0	24.5	225.0	16.9	830
VOL-16	Lower	68.08	0.67	16.42	4.64	2.20	3.38	4.76	5.32	143.0	248.0	24.9	229.0	17.2	842
VOL-17	Lower	68.26	0.69	16.54	4.73	2.25	3.46	4.82	5.39	146.0	246.0	25.3	233.0	17.5	854
VOL-18	Lower	68.44	0.71	16.66	4.82	2.30	3.54	4.88	5.46	149.0	244.0	25.7	237.0	17.8	866
VOL-19	Lower	68.62	0.73	16.78	4.91	2.35	3.62	4.94	5.53	152.0	242.0	26.1	241.0	18.1	878
VOL-20	Lower	68.80	0.75	16.90	5.00	2.40	3.70	5.00	5.60	155.0	240.0	26.5	245.0	18.4	890
VOL-21	Lower	68.98	0.77	17.02	5.09	2.45	3.78	5.06	5.67	158.0	238.0	26.9	249.0	18.7	902
VOL-22	Lower	69.16	0.79	17.14	5.18	2.50	3.86	5.12	5.74	161.0	236.0	27.3	253.0	19.0	914
VOL-23	Lower	69.34	0.81	17.26	5.27	2.55	3.94	5.18	5.81	164.0	234.0	27.7	257.0	19.3	926
VOL-24	Lower	69.52	0.83	17.38	5.36	2.60	4.02	5.24	5.88	167.0	232.0	28.1	261.0	19.6	938
VOL-25	Lower	69.70	0.85	17.50	5.45	2.65	4.10	3.80	5.95	170.0	230.0	28.5	265.0	19.9	950

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Table 4 (Continued).

Sample	Unit	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	Rb	Sr	Y	Zr	Nb	Ba
VOL-26	Lower	69.88	0.87	17.62	5.54	2.70	4.18	3.86	6.02	173.0	228.0	28.9	269.0	20.2	962
VOL-27	Lower	70.06	0.89	17.74	5.63	2.75	4.26	3.92	6.09	176.0	226.0	29.3	273.0	20.5	974
VOL-28	Lower	70.24	0.91	17.86	5.72	2.80	4.34	3.98	6.16	179.0	224.0	29.7	277.0	20.8	986
VOL-29	Lower	70.42	0.93	17.98	5.81	2.85	4.42	4.04	4.23	182.0	222.0	30.1	281.0	21.1	998
VOL-30	Lower	70.60	0.95	14.60	5.90	2.90	4.50	4.10	4.30	185.0	220.0	30.5	285.0	12.4	1010
VOL-31	Middle	70.78	0.97	14.72	5.99	2.95	4.58	4.16	4.37	188.0	218.0	30.9	289.0	12.7	1022
VOL-32	Middle	70.96	0.99	14.84	6.08	3.00	4.66	4.22	4.44	191.0	216.0	31.3	293.0	13.0	1034
VOL-33	Middle	71.14	1.01	14.96	6.17	3.05	4.74	4.28	4.51	194.0	214.0	31.7	297.0	13.3	1046
VOL-34	Middle	71.32	1.03	15.08	3.26	3.10	4.82	4.34	4.58	197.0	212.0	32.1	301.0	13.6	658
VOL-35	Middle	71.50	1.05	15.20	3.35	3.15	4.90	4.40	4.65	200.0	210.0	32.5	165.0	13.9	670
VOL-36	Middle	71.68	1.07	15.32	3.44	3.20	4.98	4.46	4.72	203.0	208.0	32.9	169.0	14.2	682
VOL-37	Middle	71.86	1.09	15.44	3.53	3.25	5.06	4.52	4.79	96.0	206.0	33.3	173.0	14.5	694
VOL-38	Middle	72.04	1.11	15.56	3.62	3.30	5.14	4.58	4.86	99.0	204.0	18.7	177.0	14.8	706
VOL-39	Middle	72.22	1.13	15.68	3.71	3.35	5.22	4.64	4.93	102.0	202.0	19.1	181.0	15.1	718
VOL-40	Middle	72.40	0.35	15.80	3.80	3.40	5.30	4.70	5.00	105.0	200.0	19.5	185.0	15.4	730
VOL-41	Middle	72.58	0.37	15.92	3.89	3.45	5.38	4.76	5.07	108.0	198.0	19.9	189.0	15.7	742
VOL-42	Middle	72.76	0.39	16.04	3.98	3.50	5.46	4.82	5.14	111.0	196.0	20.3	193.0	16.0	754
VOL-43	Middle	72.94	0.41	16.16	4.07	3.55	5.54	4.88	5.21	114.0	194.0	20.7	197.0	16.3	766
VOL-44	Middle	73.12	0.43	16.28	4.16	3.60	5.62	4.94	5.28	117.0	192.0	21.1	201.0	16.6	778
VOL-45	Middle	73.30	0.45	16.40	4.25	3.65	5.70	5.00	5.35	120.0	190.0	21.5	205.0	16.9	790
VOL-46	Middle	73.48	0.47	16.52	4.34	3.70	5.78	5.06	5.42	123.0	188.0	21.9	209.0	17.2	802
VOL-47	Middle	73.66	0.49	16.64	4.43	3.75	5.86	5.12	5.49	126.0	186.0	22.3	213.0	17.5	814
VOL-48	Middle	73.84	0.51	16.76	4.52	3.80	2.14	5.18	5.56	129.0	184.0	22.7	217.0	17.8	826
VOL-49	Middle	74.02	0.53	16.88	4.61	3.85	2.22	5.24	5.63	132.0	182.0	23.1	221.0	18.1	838
VOL-50	Middle	74.20	0.55	17.00	4.70	1.40	2.30	3.80	5.70	135.0	180.0	23.5	225.0	18.4	850
VOL-51	Middle	74.38	0.57	17.12	4.79	1.45	2.38	3.86	5.77	138.0	178.0	23.9	229.0	18.7	862
VOL-52	Middle	74.56	0.59	17.24	4.88	1.50	2.46	3.92	5.84	141.0	176.0	24.3	233.0	19.0	874
VOL-53	Middle	74.74	0.61	17.36	4.97	1.55	2.54	3.98	5.91	144.0	174.0	24.7	237.0	19.3	886
VOL-54	Middle	74.92	0.63	17.48	5.06	1.60	2.62	4.04	5.98	147.0	172.0	25.1	241.0	19.6	898
VOL-55	Middle	75.10	0.65	17.60	5.15	1.65	2.70	4.10	6.05	150.0	170.0	25.5	245.0	19.9	910

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Table 4 (Continued).

Sample	Unit	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	Rb	Sr	Y	Zr	Nb	Ba
VOL-56	Middle	75.28	0.67	17.72	5.24	1.70	2.78	4.16	6.12	153.0	168.0	25.9	249.0	20.2	922
VOL-57	Middle	75.46	0.69	17.84	5.33	1.75	2.86	4.22	6.19	156.0	166.0	26.3	253.0	20.5	934
VOL-58	Middle	75.64	0.71	17.96	5.42	1.80	2.94	4.28	4.26	159.0	164.0	26.7	257.0	20.8	946
VOL-59	Middle	75.82	0.73	14.58	5.51	1.85	3.02	4.34	4.33	162.0	162.0	27.1	261.0	21.1	958
VOL-60	Middle	76.00	0.75	14.70	5.60	1.90	3.10	4.40	4.40	165.0	160.0	27.5	265.0	12.4	970
VOL-61	Upper	76.18	0.77	14.82	5.69	1.95	3.18	4.46	4.47	168.0	158.0	27.9	269.0	12.7	982
VOL-62	Upper	76.36	0.79	14.94	5.78	2.00	3.26	4.52	4.54	171.0	156.0	28.3	273.0	13.0	994
VOL-63	Upper	76.54	0.81	15.06	5.87	2.05	3.34	4.58	4.61	174.0	154.0	28.7	277.0	13.3	1006
VOL-64	Upper	76.72	0.83	15.18	5.96	2.10	3.42	4.64	4.68	177.0	152.0	29.1	281.0	13.6	1018
VOL-65	Upper	76.90	0.85	15.30	6.05	2.15	3.50	4.70	4.75	180.0	150.0	29.5	285.0	13.9	1030
VOL-66	Upper	77.08	0.87	15.42	6.14	2.20	3.58	4.76	4.82	183.0	148.0	29.9	289.0	14.2	1042
VOL-67	Upper	65.26	0.89	15.54	3.23	2.25	3.66	4.82	4.89	186.0	146.0	30.3	293.0	14.5	654
VOL-68	Upper	65.44	0.91	15.66	3.32	2.30	3.74	4.88	4.96	189.0	144.0	30.7	297.0	14.8	666
VOL-69	Upper	65.62	0.93	15.78	3.41	2.35	3.82	4.94	5.03	192.0	142.0	31.1	301.0	15.1	678
VOL-70	Upper	65.80	0.95	15.90	3.50	2.40	3.90	5.00	5.10	195.0	140.0	31.5	165.0	15.4	690
VOL-71	Upper	65.98	0.97	16.02	3.59	2.45	3.98	5.06	5.17	198.0	138.0	31.9	169.0	15.7	702
VOL-72	Upper	66.16	0.99	16.14	3.68	2.50	4.06	5.12	5.24	201.0	136.0	32.3	173.0	16.0	714
VOL-73	Upper	66.34	1.01	16.26	3.77	2.55	4.14	5.18	5.31	204.0	134.0	32.7	177.0	16.3	726
VOL-74	Upper	66.52	1.03	16.38	3.86	2.60	4.22	5.24	5.38	97.0	132.0	33.1	181.0	16.6	738
VOL-75	Upper	66.70	1.05	16.50	3.95	2.65	4.30	3.80	5.45	100.0	280.0	18.5	185.0	16.9	750
VOL-76	Upper	66.88	1.07	16.62	4.04	2.70	4.38	3.86	5.52	103.0	278.0	18.9	189.0	17.2	762
VOL-77	Upper	67.06	1.09	16.74	4.13	2.75	4.46	3.92	5.59	106.0	276.0	19.3	193.0	17.5	774
VOL-78	Upper	67.24	1.11	16.86	4.22	2.80	4.54	3.98	5.66	109.0	274.0	19.7	197.0	17.8	786
VOL-79	Upper	67.42	1.13	16.98	4.31	2.85	4.62	4.04	5.73	112.0	272.0	20.1	201.0	18.1	798
VOL-80	Upper	67.60	0.35	17.10	4.40	2.90	4.70	4.10	5.80	115.0	270.0	20.5	205.0	18.4	810

Notes: Major oxides in wt%; trace elements in ppm. LOI measured at 1000 °C.